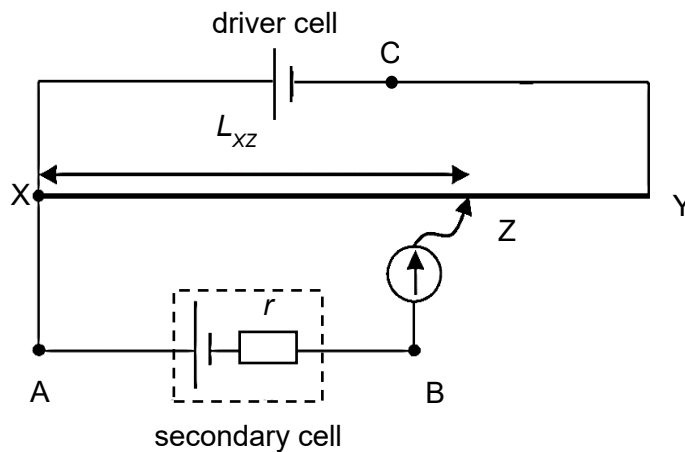


22 A typical potentiometer circuit is shown.



Which of the following does **not** increase the balance length L_{XZ} ?

- A decreasing the e.m.f. of the driver cell.
- B increasing the e.m.f. of the secondary cell.
- C adding a fixed resistor in series with the driver cell at point C.
- D decreasing the internal resistance of the secondary cell.

Ans: D

Option A: this decreases V_{XY} . L_{XZ} has to increase in order for V_{XZ} to still be equal to e.m.f of secondary cell.

Option B: this means L_{XZ} has to increase to increase V_{XZ} to match e.m.f of secondary cell.

Option C: this causes the same effect as in option A.

Option D: at balance length, the current is through secondary cell is zero, thus V_{AB} will still be equal to the e.m.f. of the secondary cell. Thus L_{XZ} does not change.

23 In the direction opposite of an electric field line, which of the following must be true?